

ANNE ARUNDEL COMMUNITY COLLEGE

Arnold, Maryland

Course Name & Number: **Calculus and Analytic Geometry 2, Math 192**
 Credit Hours: 4 credit hours (4 hours of lecture weekly)
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CATALOG DESCRIPTION

Learn numerical and analytical techniques for integration and apply these techniques to solve problems involving definite integrals. Other topics include solving separable differentiable equations, extending the concepts of calculus to polar and parametric forms, using Taylor polynomials to approximate functions, and determining the convergence or divergence of improper integrals, infinite sequences and infinite series.

Prerequisite: MAT 191 or equivalent.

Crosslisted: Also offered as MAT 192H; credit is not given for both MAT 192 and MAT 192H.

LEARNING OUTCOMES

Upon successful completion of this course, a student will be able to:

- a. Use definite integral to find the area, volume, work, arc-length, surface area, etc.
- b. Use advanced techniques of integration to evaluate definite and indefinite integrals.
- c. Solve separable differential equations, including applications.
- d. Approximate definite integrals numerically and analyze the error.
- e. Determine the convergence or divergence of improper integrals, infinite sequences and infinite series.
- f. Find Taylor Polynomials and apply them to finding approximations.
- g. Extend the concepts of calculus to curves in parametric and polar forms.

DIVISION OF SUBJECT MATTER

*All topics on the course outline are to be covered; the order given below is a suggestion.
Individual instructors may re-order topics.*

Main Topic	<u>Lecture Hours</u>
1.0 Review of Calculus 1 and Introduction to Calculus 2	2
2.0 Applications of Definite Integral	8
3.0 Techniques of Integration	12
4.0 Applications of Integration and Differential Equations	5
5.0 Infinite Series	18
6.0 Plane Curves and Polar Coordinates	7
Review and Testing, including Final Exam (2)	8
Total	60

DETAILED COURSE OUTLINE¹

	<u>Lecture Hours</u>
1.0 Review of Calculus 1 and Introduction to Calculus 2	2
2.0 Applications of Definite Integral	
2.1 Areas between Curves (6.1)	2
2.2 Volumes (6.2)	2
2.3 Volumes by Cylindrical Shells (6.3)	2
2.4 Work (6.4)	1
2.5 Average Value of a Function (6.5)	<u>1</u>
	8
3.0 Techniques of Integration	
3.1 Integration by Parts (7.1)	1
3.2 Trigonometric Integrals (7.2)	2
3.3 Trigonometric Substitutions (7.3)	1
3.4 Integration of Rational Functions by Partial Fractions (7.4)	2
3.5 Integration Using Tables and Computer Algebra Systems (7.6)	1
3.6 Approximate Integration and Error Analysis (7.7)	3
3.7 Improper Integrals (7.8)	<u>2</u>
	12
4.0 Further Applications of Integration and Differential Equations	
4.1 Arc Length (8.1)	1
4.2 Surface Area (8.2)	1
4.3 Differential Equations, Separable Equations, and Exponential Growth and Decay (9.1, 9.3, 9.4)	<u>3</u>
	5
5.0 Infinite Series	
5.1 Sequences (11.1)	2
5.2 Series (11.2)	2
5.3 The Integral Test and Estimates of Sums (11.3)	1
5.4 The Comparison Tests (11.4)	2
5.5 Alternating Series and Absolute Convergence (11.5)	1
5.6 The Ratio and Root Tests (11.6)	2
5.7 Strategy for Testing Series (11.7)	1
5.8 Power Series (11.8)	1
5.9 Representation of Functions as Power Series (11.9)	2
5.10 Taylor and Maclaurin Series (11.10)	2
5.11 Applications of Taylor Polynomials (11.11)	<u>2</u>
	18
6.0 Parametric Equations and Polar Coordinates	
6.1 Parametric Equations (10.1)	1
6.2 Calculus and Parametric Curves (10.2)	2
6.3 Polar Coordinates (10.3)	2
6.4 Areas and Lengths in Polar Coordinates (10.4)	<u>2</u>
	7
Review and Testing, including Final Exam (2)	<u>8</u>
Total	60

¹It is understood that students will be exposed to numerical and graphical methods in a variety of situations, as well as algebraic approaches.

TECHNOLOGY USE

Faculty members teaching MAT 192 are expected to introduce modern technology by computer software packages and/or graphing calculators. Highly recommended calculators are the TI-84 or TI-84+. The TI-83 or TI-83+ calculators also are sufficient. Technology should be used for classroom demonstration purposes and students are to be encouraged to utilize same by way of assignments. Students should be prohibited from using a TI-89, TI-92, TI-Inspire, or other calculator containing a Computer Algebra System (CAS) that performs symbolic operations during proctored assessments. Students may not use a TI-emulator or other graphing calculator app on their smartphones or other electronic devices during proctored assessments. Any use, during proctored assessments, of a device that does symbolic computations will not meet the objectives of the course.

TEXTBOOK

Title: Calculus: Early Transcendentals, 9th ed.
 Authors: James Stewart, Daniel Clegg, and Saleem Watson
 Publisher: Cengage
 Date: 2019

Instructors may request a physical copy of the text from the Mathematics Department. Instructors can also obtain access to the eText, WebAssign, and other resources through a Cengage Instructor username/password (see Instructor Resources section, below). For students, a physical copy of the text is available from the bookstore or directly from the publisher in several formats. Students may also purchase a one- or three-semester access code for the online homework and eText (WebAssign) through the bookstore or directly from the publisher. A student solutions manual is also available through the bookstore or directly from the publisher.

INSTRUCTOR RESOURCES

The following resources are available for use with this textbook:

- Complete solutions manual. A paper copy may be available from the mathematics department. PDF version available online from “Instructor’s Companion Site.”
- Powerpoint slides available online from “Instructor’s Companion Site.”
- Instructor’s Guide. Listed in text as available from “Instructor’s Companion Site.”
- Test Bank: text-specific multiple-choice and free response test items. Listed in text as available from “Instructor’s Companion Site.”
- Accessing the “Instructor’s Companion Site”:
 - Go to <https://www.cengage.com/dashboard/#/login> and input your Cengage instructor username and password.
 - Now that you’ve entered your Cengage Instructor Resource Account Dashboard, you can search for your title’s resources by using ISBN, author or title in the ‘Search’ box in the center of the screen. **Our text: ISBN-13: 978-1-337-61392-7**
 - You will be able to check the boxes under *Add complimentary Student Resources and all Instructor Companion Site resources to your Instructor Resource Center*. All choices include access to the “instructor’s Companion Site.”
 - Click Add Selected to Instructor Resource Center
 - If you have questions about this process or need an instructor username/password, contact Cengage Tech Support at (800-354-9706) or our Cengage Representative, Bianca Fiorio, Bianca.fiorio@cengage.com.

REQUIRED SYLLABUS STATEMENTS

There are several AACC policy statements that faculty are required to include on their syllabi each semester. The updated required statements are emailed to faculty each semester, and can also be obtained by contacting the Mathematics Department adjunct faculty coordinator. As of the writing of this document, the statements include:

- Americans with Disabilities Act policy
- academic integrity policy
- student conduct policy
- acceptable use of information technology resources policy
- notice of nondiscrimination
- statement of study time

ADDITIONAL REFERENCES

1. Calculus by Swokowski, Olinick, and Pence
2. Calculus Early Transcendentals, by Anton, Bivens and Davis
3. Calculus by Hughes-Hallett, Gleason, and McCallum.
4. Calculus, Early Transcendental Functions, by Larson, Hostetler and Edwards
5. Calculus, Early Transcendental Functions, by Smith and Minton
6. Thomas' Calculus: Early Transcendentals by Hass, Heil, and Weir.